AI Vision Extract - Technical Documentation

1. Project Overview

AI Vision Extract automatically isolates the main subject from any input image, producing clean cutouts with customizable backgrounds (black, white, studio gray).

Core Pipeline:

```
Input Image → DeepLabV3 Segmentation → Binary Subject Mask → Subject Extraction → Post-processing → Output
```

Key Performance Metrics:

Model	Mean IoU	Global Pixelwise Accuracy
DeepLabV3 ResNet101	67.4%	92.4%

Applications: Photography automation, e-commerce, AR/VR backgrounds, virtual conferencing.

2. COCO 2017 Dataset Processing

Dataset: 118K train + 5K val images, 80 object categories with instance masks.

Data Quality Pipeline (99% Valid Retention)

```
1. Anomaly Detection: Missing files, corrupted images, tiny objects (<32px), invalid masks
2. Semantic Mask Extraction: Multi-class → Category ID per pixel (uint16 PNGs)
3. Output: 117,254 train masks + 4,952 val masks = 122,206 total clean masks
```

Validation Results:

- Train: 99.1% valid (117,254/118,287)
- Val: 99.0% valid (4,952/5,000)

3. Model Architecture & Training

DeepLabV3 ResNet101 Configuration

```
Backbone: ResNet101 (101-layer feature extractor)
ASPP: Atrous Spatial Pyramid Pooling (multi-scale context)
Output: 81 channels (80 COCO classes + background)
Loss: CrossEntropyLoss (multi-class segmentation)
```

Training Setup:

- **Optimizer:** SGD (lr=0.01, momentum=0.9, weight_decay=1e-4)
- **Batch Size:** 8 (GPU-optimized)
- **Input:** 520×520, ImageNet normalized (μ=[0.485,0.456,0.406], σ=[0.229,0.224,0.225])
- **Results:** mIoU 67.4%, Pixel Accuracy 92.4%

4. Preprocessing Pipeline

Training Data Preparation

```
COCOSegmentationDataset:
├── Images: JPEG → Resize(520×520) → Normalize → Tensor
├── Masks: PNG(uint16) → torch.long tensor (category IDs 0-80)
└── DataLoader: batch_size=8, shuffle=True
```

Inference Pipeline

```
1. Resize(520) → Normalize → Model Inference
2. argmax(81) → Binary mask (>0 = subject)
3. Resize to original → Apply background fill
4. Brightness/Contrast → Optional auto-crop → PNG export
```

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5. Streamlit Production App

Interactive Features:

```
Upload: Multiple JPG/PNG/JPEG files
Controls: Background (Black/White/Gray), Brightness(0.5-2.0), Contrast(0.5-2.0), Auto-crop
Metrics: Subject coverage %, original resolution
Export: Individual PNGs + Batch ZIP archive
```

UI Layout:

```
Sidebar: Controls        Main: Original | Result (side-by-side)
Metrics: Coverage 92%    Res: 1920×1080  Download PNG/ZIP
```

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6. Technical Results Summary

Metric	Value	Description
Mean IoU	**67.4%**	Overlap between predicted/ground-truth masks
Pixel Accuracy	**92.4%**	Correctly classified pixels
Train Masks	117,254	Clean semantic masks generated
Val Masks	4,952	Validation set quality
Data Purity	99.0%	Post-anomaly filtering retention

Validation: High pixel accuracy confirms robust subject-background separation across diverse scenes.

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7. Future Enhancements

1. **Domain Fine-tuning:** E-commerce products, portrait optimization
2. **Video Processing:** Real-time frame-by-frame segmentation
3. **Model Optimization:** ONNX export, INT8 quantization, edge deployment
4. **Advanced UI:** Multi-subject extraction, interactive mask refinement

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